# FoveaWAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

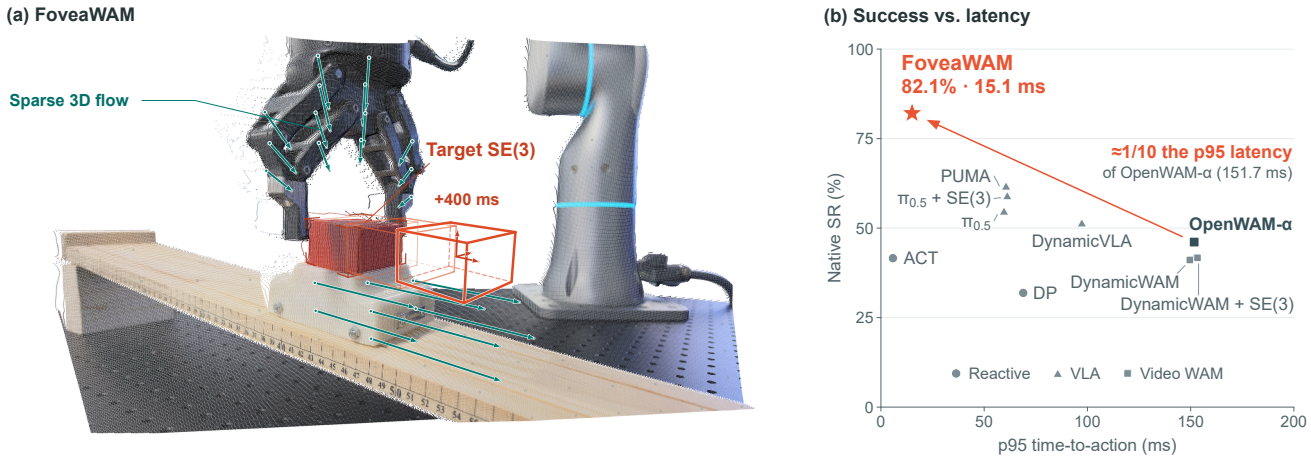



Figure 1. **FoveaWAM allocates prediction precision by task role.** (a) A precise SE(3) trajectory for the target and sparse 3D flow for its surroundings. (b) SR versus p95 time-to-action on OnTimeBench.

## Abstract

*To manipulate a moving object, a policy must predict the scene at the moment its action executes, yet every millisecond spent predicting pushes that moment further away. Existing policies either forecast the whole scene at uniform fidelity and act late, or act fast by ignoring everything but the target. Our key insight is that what a policy must predict decouples by role: only the target's pose, which decides contact, needs millimeter precision, while the surroundings matter only through the clearance they leave and the motion they impart. We present FoveaWAM, a native 4D world–action model that allocates precision by role: it models the target's past and future motion as a metric SE(3) trajectory and the surroundings as sparse 3D flow, and relies on the current frame for everything static. This compact representation lets FoveaWAM act in 15.1 ms on one RTX 5090, about* $10\times$ *faster than video world–action baselines, giving the scene little time to move. To evaluate each policy under its own latency, we introduce OnTimeBench, a dynamic manipulation benchmark in which the scene keeps moving while the policy computes. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline, and also leads on DOMINO and real-robot tasks.*

## 1. Introduction

To manipulate a moving object, a robot must act on a scene it has not yet seen. Objects keep moving while the policy perceives, computes, and acts, so every action lands in a later scene than the one observed. A ball rolling down a ramp at about 0.9 m/s moves 18 cm during a 200 ms inference; contact tolerates millimeters. The policy must therefore predict where the scene will be when its action executes. But prediction is itself computation, and its cost lengthens the interval it must cover: the longer a policy computes to anticipate motion, the further ahead it must anticipate. Existing dynamic policies settle this tension at one of two ends (Fig. 2). Some forecast the whole scene at uniform fidelity, as future video, latents, or 3D dynamics [29, 30, 44, 49, 54, 57]; they capture everything but act late. Others forecast only the manipulated object [14, 17]; they act sooner but miss the surroundings that block or redirect it. Either way, every part of the scene that is predicted costs the same, so knowing more means acting later. We ask instead: *what must a policy know about a moving scene, and how precisely, to make contact in time?*

Our key insight is that the answer decouples by role (Fig. 1). The target decides contact, so its pose at the moment of execution must be known to millimeter precision. The surroundings matter only through the clearance they leave

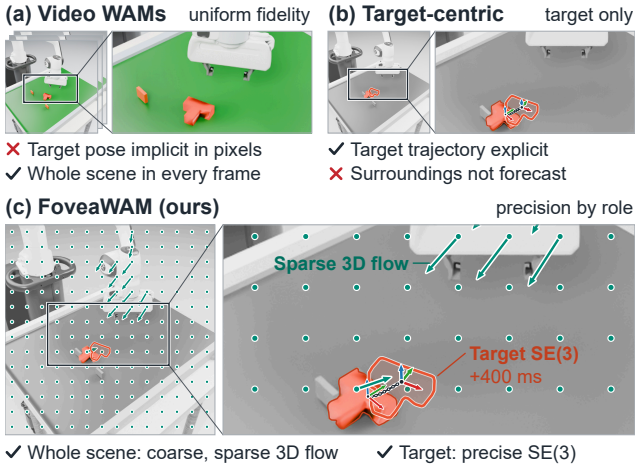



Figure 2. **Where forecast precision goes.** (a) Video WAMs forecast the whole scene and leave the target pose implicit [44, 49]. (b) Target-centric predictors forecast only the target's trajectory [14, 17]. (c) FoveaWAM forecasts a precise SE(3) trajectory for the target and sparse 3D flow for the scene.

and the motion they impart to the target, so their motion need only be known coarsely. Everything static is already in the current frame and needs no prediction at all. What must be known precisely is therefore one object's trajectory, not the scene, and a policy that spends its precision there can be exact where contact is decided and still act in time. Biological vision makes the same allocation: the fovea resolves fine detail only where gaze lands, the periphery stays coarse yet alert to motion, and gaze runs ahead of the hand to where contact will occur [12, 19, 23, 33, 38].

We build this allocation into **FoveaWAM**, a native 4D world–action model whose representation follows the three roles (Fig. 2c). The target is described by a metric SE(3) trajectory: a short pose history supplies its past motion in place of past image frames, and the model learns to forecast its continuation. The surroundings are described by sparse 3D flow, and everything static by the current frame alone. Both forecasts supervise the representation during training and are dropped at inference [53]. As a result, FoveaWAM acts in 15.1 ms on one RTX 5090, about 10× faster than video world–action models, leaving the scene little time to move.

Measuring this advantage requires an evaluation in which latency has a cost. The default evaluators of DOMINO and PhysMani-Bench pause the world while the policy computes [14, 54], so a slow policy is scored as if it were instant. We introduce **OnTimeBench**, 14 dynamic manipulation tasks in which the world never pauses: the scene keeps moving for as long as each policy computes, and every method is evaluated at its native latency. The change of protocol reorders existing policies: DynamicWAM with SE(3) history beats the lightweight ACT [59] by 23.4 points with the world paused, but by only 0.1 points at native latency. FoveaWAM reaches 82.1% success on OnTimeBench, 20.7 points above the strongest baseline. Its lead holds with the world paused and on DOMINO [14], and it succeeds in 37.5% of trials across six real-robot tasks. Ablations support the allocation: coarsening the target's forecast or removing its pose history lowers success. Uniform fine flow recovers overall success but trails on both the rotation-sensitive and context-active task groups.

Our contributions are threefold:

- We show that what a dynamic policy must know decouples by role: millimeter precision for the target's pose at execution time, coarse motion for the surroundings, and no prediction for the static scene.
- We present **FoveaWAM**, a native 4D world–action model that allocates precision by role and acts in 15.1 ms, about 10× faster than video world–action models.
- We introduce **OnTimeBench**, a dynamic manipulation benchmark that evaluates each policy at its native latency and reorders existing policies relative to paused evaluation. FoveaWAM reaches 82.1% success on it, 20.7 points above the strongest baseline, and also leads on DOMINO.

## 2. Related Work

**Dynamic manipulation.** Acting on moving objects, long handled by engineered tracking and interception [1–3, 32, 36], is now learned from data [9, 25, 47]. Learned policies take motion from image-space cues, such as optical flow in PUMA [14] and DynamicWAM [29], or act on test-time forecasts of the scene [30, 40, 54, 57]. FoveaWAM instead reads the target's metric SE(3) pose history and observes only the current frame. Execution schemes [6, 27, 35, 41, 47] change when actions are applied, not what the policy represents, and are complementary. DOMINO [14] and PhysMani-Bench [54] provide diverse dynamic tasks but by default pause the simulation until the policy returns; OnTimeBench advances it during inference, so each policy is evaluated at its own latency.

**Forecasting in world–action models.** WAMs generate future video and actions in one network [24, 44, 49, 61], building on video prediction as a policy objective [42, 46]. Because video is costly to generate, recent work forecasts cheaper targets, such as features, flow, object slots, or compact latents [28, 39, 50, 52, 56, 60], and some use the forecast only in training [14, 53, 58]. In all of them the forecast lives in pixel, feature, or latent space, and the target's future pose stays implicit. FoveaWAM keeps the training-only use but forecasts a metric 4D state whose precision follows role, so the target's future pose is supervised explicitly.

**Object-centric and geometric motion representations.** Metric 3D inputs improve precision and sample efficiency [34, 55], and motion has been represented by point tracks [4, 45, 48], 3D flow [15, 18, 51, 54], object keypoints or SE(3) trajectories [17, 31, 37], and rigid-body decompositions [7, 8]. Each applies one representation to the whole scene, or models the target alone [14, 17]. DOCIR separates robot, target, and obstacles, but only as input encodings [13], and foveated robot perception allocates image resolution around a gaze point [11, 20, 22]. FoveaWAM allocates the precision of what is predicted: coarse sparse flow for the scene and a metric SE(3) trajectory for the target.

# 3. FoveaWAM

We present FoveaWAM, a native 4D world–action model that allocates precision by role on both sides of the present (Fig. 3). It encodes the past (Sec. 3.2) and predicts the future (Sec. 3.3) with the same split: a metric SE(3) trajectory for the target, coarse 3D geometry for its surroundings, and the current frame alone for everything static. An action stream reads this representation to generate time-indexed commands (Sec. 3.4).

## 3.1. Problem Formulation

We study manipulation in scenes that keep moving while the policy computes. At query $t$, a policy $\pi$ learned from demonstrations reads an RGB image $I_t$ and a depth map $D_t$ captured at time $\tau_t$, together with histories $\mathcal{T}_t$ of the target pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ and $\mathcal{R}_t$ of the robot state, all in the robot base frame, and outputs a chunk of $H$ commands

$$(\mathbf{a}_{t,0}, \dots, \mathbf{a}_{t,H-1}) \sim \pi\big(\cdot \mid I_t, D_t, \mathcal{T}_t, \mathcal{R}_t\big), \qquad (1)$$

where command $j$ is meant for time $\tau_t + j\Delta_a$, with control period $\Delta_a$.

Each task assigns the scene three roles. The *target* is the object whose pose the robot must match within a contact or containment tolerance. The *context* is the part of the surroundings that matters: obstacles, which the robot must clear, and drivers, which move the target. Everything else is *static*. The chunk becomes available only after inference, so each executed command must account for the target and the context at its execution time rather than at $\tau_t$.

## 3.2. Role-Aware Dynamics Encoding

What a policy must know about the past decouples by role. The static scene has no history worth encoding: the current frame already gives its layout. The context acts through the clearance it leaves, which is current geometry, and the motion it imparts, which shows in how the target has moved. Only the target's motion must be known precisely, and that motion is a rigid trajectory in SE(3). FoveaWAM therefore observes a single RGB-D frame for the scene and a short metric pose history for the target. Policies that take motion from stacked frames [10, 47] or optical-flow histories [14, 29] extract it densely over the whole image at every past step, and the target's velocity still remains implicit in appearance. FoveaWAM receives that velocity in physical units from a few low-dimensional tokens. This encoding keeps the model light enough to act within one control period while keeping the target's motion metric.

**Static scene from the current frame.** FoveaWAM anchors each patch token at a metric 3D point, so that the observation shares the base frame of the predictions and the actions. A frozen MAE ViT-B/16 [16] turns the RGB image into 196 patch tokens on a $14 \times 14$ grid, and the depth map back-projects the center of patch $i$ to a point $\mathbf{x}_i \in \mathbb{R}^3$ in the base frame. A linear embedding of $\mathbf{x}_i$ is added to the patch token, and $\mathbf{x}_i$ drives 3D rotary position embeddings of the patch queries and keys, so that attention between patches depends on their relative 3D positions. One frame thus gives the scene layout and its clearances as metric geometry. The points $\mathbf{x}_i$ are also the source points of the scene flow (Sec. 3.3).

**Target motion from the pose history.** The target's recent motion enters as a metric SE(3) state. The history spans the last 350 ms in 8 samples at 20 Hz; its 8 tokens hold the current target pose and the seven motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units. The velocity and acceleration that decide where the target will be thus enter the model directly, and a driver such as a conveyor or a ramp is captured through the motion it has imparted to the target. Poses come from the simulator in simulation and from an online RGB-D tracker on the robot.

**World stream.** The patch tokens, the target tokens, and 8 robot tokens with the end-effector states at the times of the target history form the 212 scene tokens. Together with 16 learnable summary tokens that aggregate them, they make up the world stream. The entire motion history thus costs sixteen low-dimensional tokens, whereas each additional frame would cost 196.

## 3.3. Role-Aware 4D Future Prediction

Every command lands in a later scene than the one observed, so the representation must anticipate the scene at the execution times of the commands, and the precision this requires again follows role. Contact is decided by the target's pose, so FoveaWAM predicts the target's full SE(3) trajectory. The context matters through clearance and imparted motion, so FoveaWAM predicts the motion of the scene only coarsely,

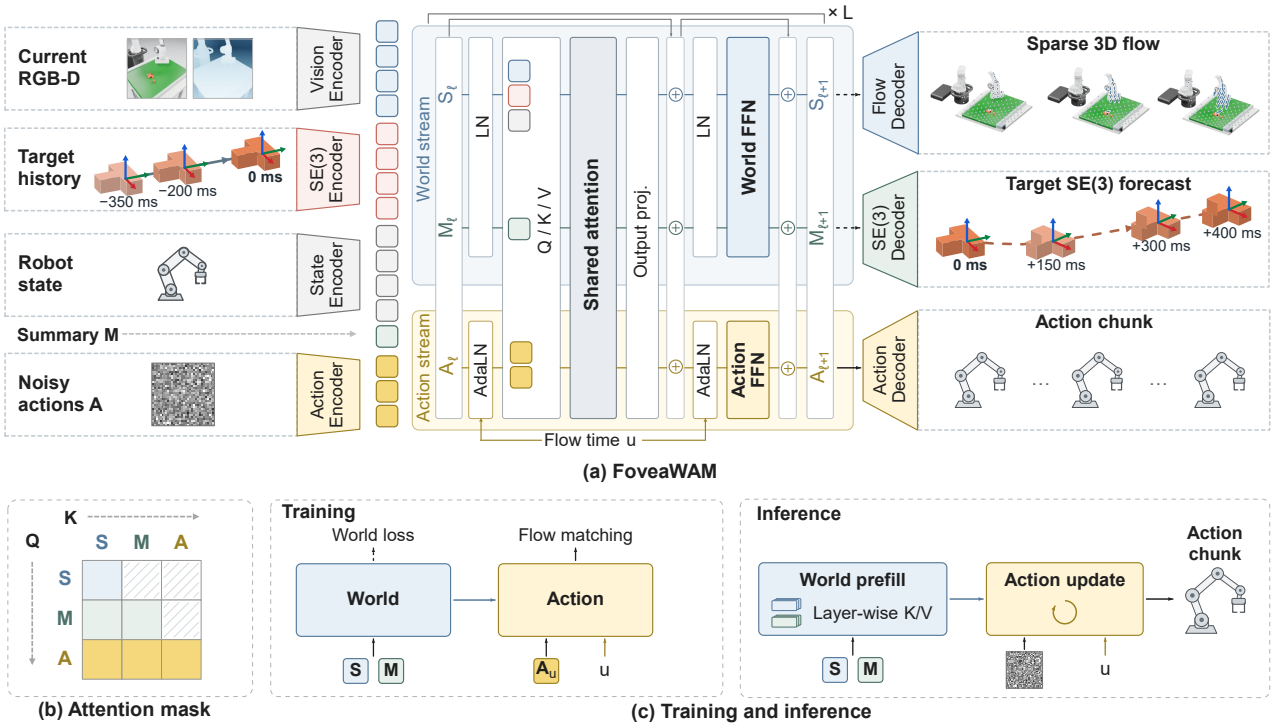



Figure 3. **Overview of FoveaWAM.** (a) Architecture. The current RGB-D frame and the target and robot histories form scene tokens $S$, alongside summary tokens $M$ and noisy action tokens $A$. All tokens share masked attention, followed by separate world and action FFNs. The sparse 3D flow and the target SE(3) trajectory are predicted from $S$ and $M$. (b) Attention mask. Rows are queries and columns are keys. (c) Training and inference. The world and action streams are trained jointly. At inference, the world stream runs once and the prediction heads are skipped.

as a sparse 3D flow on the patch grid. Both predictions are expressed in the base frame as changes from the current state at the times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50\,\text{ms}$ for $h = 1, \ldots, 8$; together with the current state, these horizons cover the execution times of all commands in the chunk. Video and latent forecasts [44, 49] spend the same capacity on every part of the scene and leave the target's pose implicit. Our two predictions are linear readouts of tokens that the action stream already reads: they shape the representation during training and are skipped at inference, so prediction adds no latency.

**Sparse 3D scene flow.** At the resolution of the patch grid, the flow covers every surface that the patches see: the robot, the target, and the context, whose clearance and motion the commands must account for. For each of the 196 points $\mathbf{x}_i$ and each horizon $h$, a linear head on the final state of patch token $i$ predicts the point's base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, trained with

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h)\in\mathcal{V}_F} \Big[ \tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{2}$$

where $F_{ih}$ is the displacement of the surface point at $\mathbf{x}_i$, $o_{ih}$ its visibility, $s$ a fixed scale fitted on the training labels, and $\mathcal{V}_F$ the set of valid labels.

**Target SE(3) trajectory.** The flow samples the target only at the few patches it covers, too sparse to determine its rotation. A dedicated linear head on the mean of the final summary states therefore predicts its rigid motion directly: for each horizon $h$, the target's displacement $\hat{\mathbf{d}}_h$ and a rotation increment $\hat{\phi}_h$, trained with

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h\in\mathcal{V}} \Big[ \|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{3}$$

where $\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t)$ and $R_h = R(\tau_t{+}\delta_h)$ are the true displacement and orientation, $\hat{R}_h = \mathrm{Exp}(\hat{\phi}_h^{\wedge}) R(\tau_t)$,

$d_{\mathrm{SO}(3)}$ is the geodesic distance, and $\mathcal{V}$ is the set of horizons with valid labels. Contact demands the rotation term: a rotation error $\theta$ displaces a contact point at distance $r$ from the target's origin by up to $2r\sin(\theta/2)$, nearly 9 mm for $5^\circ$ at $r = 10$ cm.

### 3.4. World–Action Learning and Inference

**Action stream.** The action stream generates the chunk by conditional flow matching [26]. It holds $H = 8$ tokens, each embedding one noisy command: an end-effector translation increment, a rotation increment, and a gripper command. In every layer, one attention spans both streams with shared projections, followed by separate feed-forward networks; world tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time. With $S$, $M$, and $A$ denoting the scene, summary, and action tokens, the attention mask allows $S \to S$, $M \to \{S, M\}$, and $A \to \{S, M, A\}$, where $X \to \mathcal{Y}$ means that tokens in $X$ attend to tokens in $\mathcal{Y}$. The action stream thus reads, in every layer, both the patch states shaped by the scene flow and the summary states shaped by the target trajectory, while the world stream never reads the actions.

**Training objective.** For a chunk $\mathbf{a}_1$ normalized with training statistics, noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream $v_\psi$ regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E}\,\mathrm{MSE}\big(v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t),\, \mathbf{a}_1 - \mathbf{a}_0\big), \tag{4}$$

where $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, $\mathcal{S}_t$ denotes the inputs in Eq. (1), and the MSE averages over valid commands and action dimensions. The world and action streams are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_{\mathrm{aux}} \mathcal{L}_{\mathrm{aux}}, \tag{5}$$

where $\mathcal{L}_F$ and $\mathcal{L}_P$ are the flow and pose losses of Eqs. (2) and (3), and $\mathcal{L}_{\mathrm{aux}}$ sums auxiliary appearance losses.

**Inference.** At deployment, FoveaWAM skips both prediction heads [24, 53, 56]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on the cached keys and values of the scene and summary tokens. One query takes 15.1 ms at p95 on one RTX 5090, within a single control period. Commands are indexed by time from capture, so the policy needs no latency input: the executor hands over each chunk at the first control-period boundary at or after inference completes and starts from the command indexed to that boundary. Further implementation details are in the supplementary material.

## 4. The OnTimeBench Benchmark

### 4.1. Overview

Existing dynamic manipulation benchmarks stop the world while the policy thinks. The default evaluators of DOMINO [14] and PhysMani-Bench [54] advance the simulation only after the policy has returned its actions, so a slow policy is scored as if it were instant. Such evaluation measures what a policy predicts but not whether the prediction arrives in time, which is what decides success on a real robot.

To evaluate whether a policy can act on a moving scene in time, we introduce OnTimeBench, a benchmark of 14 dynamic manipulation tasks in which the world never pauses. Observation, scene motion, and action execution share one physical clock, so the scene keeps moving for as long as each policy computes. As shown in Fig. 4(a), a rolling ball travels 18 cm during a 200 ms inference, and the policy must act on where the ball will be. OnTimeBench covers five task families (Fig. 4(b)): *Grasp*, grasping a target under harmonic, conveyor, translating, rotating, or occluded motion; *Place*, placing a held object into a translating or rotating receptacle; *Pass*, passing a wand along a rotating wire; *Capture*, capturing a rolling or sliding object; and *Impact*, grasping a part after an impact. Fig. 4(c) shows all 14 tasks. Together they vary what decides success: the target's speed, from 0.03 to 1.14 m/s, its rotation, the clearance of obstacles, and the motion imparted by drivers such as conveyors, ramps, and strikers.

### 4.2. Construction

OnTimeBench is built in MuJoCo and robosuite [43, 62] with a Franka Panda arm controlled at 20 Hz. Each scene samples the target's initial state and motion parameters within task-specific ranges. A task-specific expert with privileged simulator state provides 800 demonstrations per task, 11,200 in total. All results are reported on 1,400 held-out test scenes, 100 per task, shared by all methods; configurations are selected on separate development scenes.

### 4.3. Evaluation Protocol

**Native latency.** During each query, physics advances by the method's own inference latency, drawn from its latency distribution measured at batch size 1 on one RTX 5090. All methods share one executor, which hands over each chunk at the first control-period boundary after inference completes and executes it from the command indexed to that boundary.

**Settings and metrics.** We evaluate three settings: *Native*, at each method's own latency; *+150 ms*, with 150 ms added to every method; and *Paused*, with physics paused during inference as in existing benchmarks. We report the task-balanced success rate (SR) and the p95 time-to-action, the

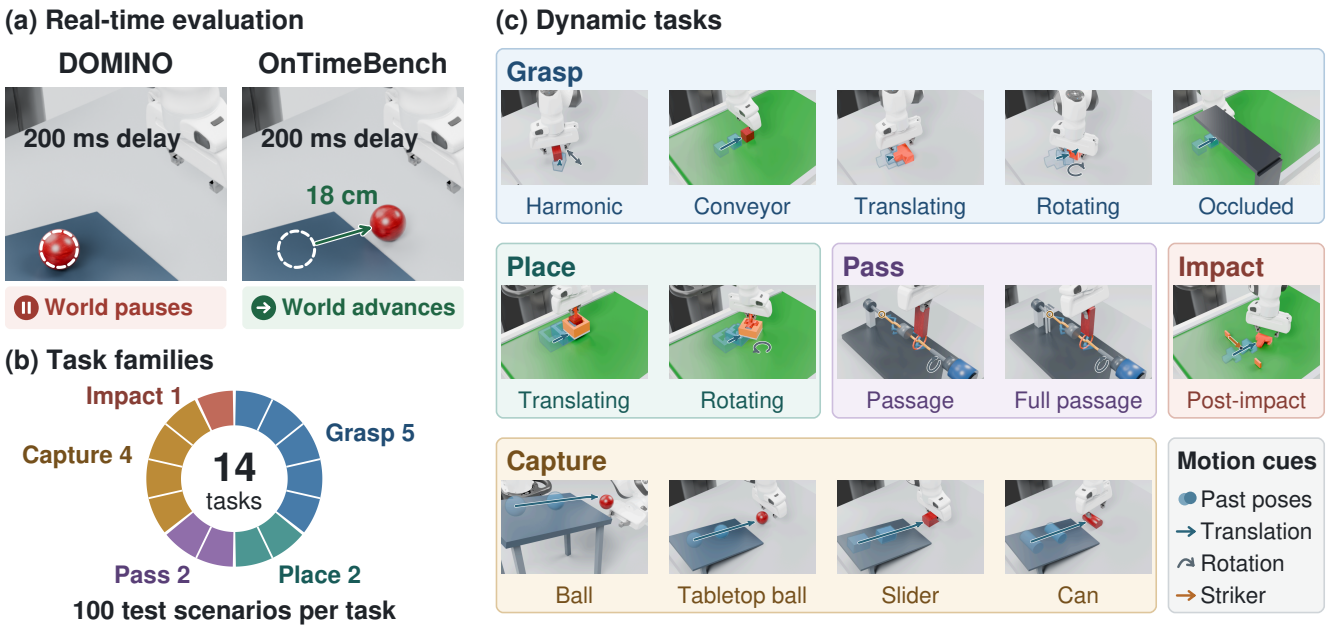



Figure 4. **Overview of OnTimeBench.** OnTimeBench evaluates each policy at its native latency, with 14 dynamic tasks across five families and 1,400 held-out test scenes. (a) Unlike DOMINO [14], which pauses the world during inference, OnTimeBench keeps it moving. (b) Distribution of task families. (c) Task demonstration.

time from an observation to its first command. The full protocol is in the supplementary material.

# 5. Experiments

## 5.1. Implementation Details

**Baselines.** We compare with the reactive policies ACT [59] and Diffusion Policy (DP) [10], the VLA $\pi_{0.5}$ [5], the dynamic VLAs DynamicVLA [47] and PUMA [14], the WAM OpenWAM-$\alpha$ [44], and the dynamic WAM DynamicWAM [29]. Each baseline keeps its official architecture and training recipe and is trained on the same demonstrations. To isolate the effect of target-motion input, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

**Model and training.** The transformer of FoveaWAM has 12 layers of width 1024 with 16 heads and 271.8M trainable parameters on top of the frozen MAE ViT-B/16 encoder. On OnTimeBench, it is trained from random initialization with AdamW for 10 epochs (25,870 updates). Inference runs 5 Euler steps in compiled BF16. FoveaWAM observes one agent-view RGB-D frame and the target's SE(3) pose history from the simulator.

## 5.2. Dynamic Manipulation Results

**Results on OnTimeBench.** As shown in Table 1, FoveaWAM achieves 82.1% SR at native latency, 20.7 points above the strongest baseline, PUMA (61.4%), and the highest SR in all three settings. It has the highest or tied-highest SR on 13 of the 14 tasks (Table S4), with the largest margins where a moving receptacle or the target's orientation decides success: 30 and 55 points on placement into translating and rotating receptacles, and 28 points on full wire passage. Precise target motion and low latency are both

Table 1. **Quantitative comparison on OnTimeBench.** SR (%) at each method's native latency, with 150 ms added, and with the world paused during inference. p95: time-to-action on one RTX 5090. History: inputs that carry past motion. (off): caching and compilation disabled. Best and second best are highlighted.

| Method | History | | | SR (%) ↑ | | | p95 (ms) ↓ |
|---|---|---|---|---|---|---|---|
| | Wrist RGB | Flow | SE(3) | Native | +150 ms | Paused | |
| ACT [59] | | | | 41.6 | 24.6 | 40.6 | **5.8** |
| DP [10] | ✓ | | | 31.9 | 16.1 | 32.0 | 68.9 |
| $\pi_{0.5}$ [5] | | | | 54.4 | 28.1 | 66.6 | 59.6 |
| $\pi_{0.5}$+SE(3) | | | ✓ | 58.7 | 28.0 | 76.0 | 61.2 |
| DynamicVLA [47] | ✓ | | | 51.1 | 23.5 | 54.9 | 97.3 |
| PUMA [14] | | ✓ | | 61.4 | 31.1 | 53.1 | 60.6 |
| OpenWAM-$\alpha$ [44] | | | | 46.1 | 24.8 | 58.7 | 151.7 |
| OpenWAM-$\alpha$ (off) | | | | 15.8 | 10.4 | 58.4 | 466.3 |
| DynamicWAM [29] | | ✓ | | 41.1 | 20.6 | 49.6 | 149.6 |
| DynamicWAM+SE(3) | | ✓ | ✓ | 41.7 | 24.9 | 64.0 | 153.3 |
| FoveaWAM | | | ✓ | **82.1** | **34.1** | **90.3** | 15.1 |

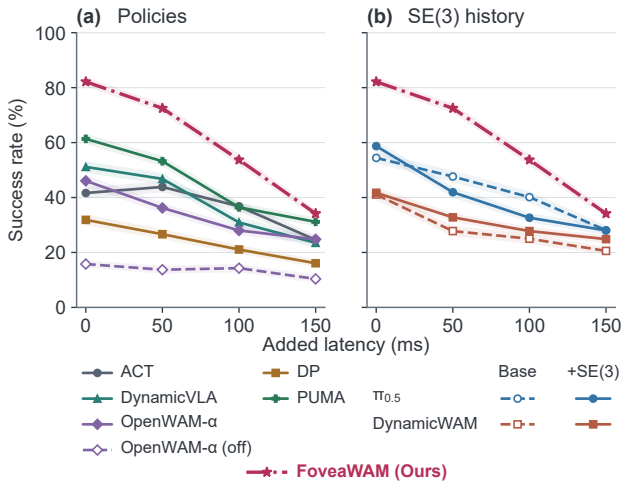



Figure 5. **Success under added latency.** SR on OnTimeBench as delay is added to each method's native latency. (a) Baselines. (off): caching and compilation disabled. (b) Base versus +SE(3) variants. Bands: 95% intervals.

required for this result. Given the same SE(3) history, $\pi_{0.5}$ and DynamicWAM reach only 58.7% and 41.7%, and ACT, the only faster method, reaches 41.6%.

**Effect of inference latency.** Latency alone can decide success. The two OpenWAM-$\alpha$ deployments run the same policy and agree with the world paused (58.7% and 58.4%), yet at native latency the one with a p95 of 151.7 ms reaches 46.1% and the one with 466.3 ms reaches 15.8%. Latency also erodes the value of motion input: the SE(3) history raises $\pi_{0.5}$ and DynamicWAM by 9.4 and 14.4 points with the world paused but by only 4.3 and 0.6 at native latency. Paused evaluation therefore overestimates slow policies and misorders methods: $\pi_{0.5}$+SE(3), the strongest baseline with the world paused (76.0%), loses 17.3 points at native latency and falls behind PUMA. FoveaWAM leads in both settings and remains the best method at every added delay (Fig. 5).

**Results on DOMINO.** To evaluate FoveaWAM beyond OnTimeBench, we train FoveaWAM, $\pi_{0.5}$, DynamicWAM, and PUMA on the 24 single-arm Franka tasks of DOMINO [14] with its demonstrations and its native evaluator. As shown in Table 2, FoveaWAM achieves the highest overall SR, 46.8%, 19.9 points above the strongest baseline, DynamicWAM (26.9%), and the highest manipulation score. This evaluator pauses physics during inference, so the lead reflects the quality of the role-aware representation rather than its speed. The margin is largest on the 16 Track tasks, 55.5% against 31.9% for DynamicWAM.

Table 2. **Quantitative comparison on DOMINO** [14]. SR (%) on its 24 single-arm tasks, 8 Intercept and 16 Track. MS: manipulation score. Best and second best are highlighted.

| Method | Intercept | Track | All ↑ | MS ↑ |
|---|---|---|---|---|
| $\pi_{0.5}$ | 23.8 | 8.1 | 13.3 | 27.8 |
| DynamicWAM | 16.8 | 31.9 | 26.9 | 41.4 |
| PUMA | **36.3** | 12.2 | 20.2 | 35.4 |
| FoveaWAM | 29.4 | **55.5** | **46.8** | **62.0** |

Table 3. **Ablation results on OnTimeBench.** SR (%) of FoveaWAM and its change for each variant ($\Delta$, points; red: drop, green: gain). Rot./Ctx.: rotation-sensitive and context-active tasks.

| | Success rate, all tasks | | Groups (Native) | |
|---|---|---|---|---|
| | Native | +150 ms | Rot. | Ctx. |
| **FoveaWAM** | **82.1** | **34.1** | **80.4** | **82.7** |
| *Precision allocation* | | | | |
| Uniform coarse flow | −4.5 | −0.5 | −8.6 | −4.6 |
| Uniform fine flow | +0.8 | −0.9 | −3.4 | −2.6 |
| w/o flow | −4.2 | −1.4 | −11.7 | −3.9 |
| *Target motion history* | | | | |
| w/o history | −19.6 | +0.2 | −24.0 | −2.3 |
| *Future supervision* | | | | |
| w/o future | −4.3 | −3.4 | −12.6 | −2.1 |
| w/o $\mathcal{L}_{\text{aux}}$ | −2.6 | −1.9 | −11.6 | +0.1 |

Table 4. **Real-robot results.** SR (%) over 20 trials per task. p95: end-to-end time-to-action (ms). Best is highlighted.

| Method | Belt grasp | Box place | Turntable | Toy car grasp | Tennis ball | Ramp grasp | All ↑ | p95 (ms) ↓ |
|---|---|---|---|---|---|---|---|---|
| $\pi_{0.5}$ | — | — | — | — | — | — | — | — |
| DynamicWAM | — | — | — | — | — | — | — | — |
| ACT | — | — | — | — | — | — | — | — |
| FoveaWAM | — | — | — | — | — | — | 37.5 | — |

## 5.3. Ablation Study

We ablate each part of the role-aware design on OnTimeBench (Table 3) and report SR at native latency, overall and on the rotation-sensitive (Rot.) and context-active (Ctx.) task groups (Table S2). All variants except *w/o history* change only the training supervision, so they run at the same latency as FoveaWAM.

**Effect of role-aware prediction.** Role-aware prediction gives the target a precision that uniform prediction does not. Replacing the SE(3) trajectory with uniform flow at the same density reduces SR by 4.5 points overall and by 8.6 on rotation-sensitive tasks. Raising the uniform flow to a $56 \times 56$ grid, $16\times$ the points, costs [X.X]$\times$ the training time of FoveaWAM; it recovers the overall SR (+0.8) but still trails by 3.4 points on rotation-sensitive tasks and by 2.6

on context-active tasks. Removing the sparse flow instead reduces SR by 4.2 points overall, by 3.9 on context-active tasks, and by 11.7 on rotation-sensitive tasks. The two predictions are thus complementary: one target trajectory and 196 flow points match 3,136 uniform flow points overall and surpass them where rotation and context matter, at a fraction of the training cost.

**Effect of target motion history.** Removing the pose history while keeping the current target pose reduces SR by 19.6 points overall and by 24.0 on rotation-sensitive tasks, the largest drop among all variants. A single pose does not show how the target moves; the metric history supplies the motion that the commands must anticipate.

**Effect of future supervision.** Supervising only the current target state instead of its future reduces SR by 4.3 points overall and by 12.6 on rotation-sensitive tasks, although the prediction heads never run at deployment. Removing the auxiliary objectives $\mathcal{L}_{\text{aux}}$ reduces SR by 2.6 points overall and by 11.6 on rotation-sensitive tasks.

### 5.4. Real-Robot Evaluation

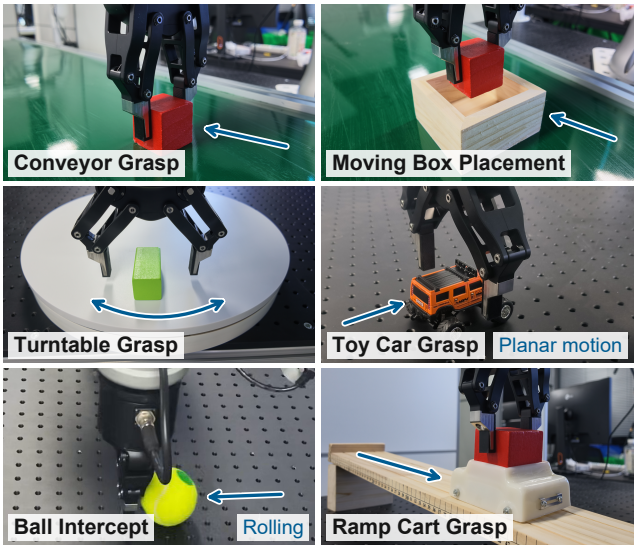



Figure 6. **Real-robot tasks.** Arrows show the motion of the target or the moving box before contact.

**Setup.** We evaluate on six dynamic tasks with a Flexiv Rizon 4s arm with a GN01 gripper (Fig. 6 and Table S8). All methods are trained on a shared set of 600 teleoperated demonstrations and evaluated for 20 trials per task. FoveaWAM starts from a DROID-pretrained base [21] and uses RGB-D and online target pose estimates from one fixed camera. The baselines, $\pi_{0.5}$, ACT, and DynamicWAM, use RGB from the fixed and wrist cameras.

**Results.** As shown in Table 4, FoveaWAM succeeds in 37.5% of all trials, against [XX.X]% for [the strongest baseline], with the largest margin on [tasks]. Its p95 latency, including tracking, is [XX] ms, against [XX] ms for [name]. [One trajectory-verified contact example: how the target's pose, rotation, or context explains a success of FoveaWAM or a failure of a baseline.]

## 6. Conclusion

We introduce FoveaWAM, a native 4D world–action model that addresses a tension at the core of dynamic manipulation: predicting a moving scene takes time, and that time lets the scene move further. By allocating precision by role, with a metric SE(3) trajectory for the target, sparse 3D flow for its surroundings, and the current frame for everything static, FoveaWAM is precise where contact is decided and acts in 15.1 ms. Together with OnTimeBench, which evaluates every policy at its native latency, it offers a model and a benchmark for building and evaluating policies that act on time.

## LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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